ICT/CE/BNT/1148

To Whom So Ever It May Concern

Subject: Clarification Regarding the Storage and Uses of n-Pentane

This report contains findings, storage, applications and recommendations with respect to n-pentane. Pentane is a colourless, volatile and flammable liquid with a sweet and gasoline-like odour. Pentane consists of three isomers: n-pentane (the most important isomer), iso-pentane, and neo-pentane. n-pentane is a constituent of crude oil and a component of the condensate from natural gas production. It is primarily obtained by fractional distillation of a petroleum stream (generally light virgin naphtha) obtained from the processing of crude oil.

n- Pentane has several uses to its credit. It is mainly used as an aerosol propellant and in low temperature thermometers. It is most preferred use is as blowing agent for foams (for example, expandable polystyrene (EPS) and polyurethane (PU foam)). In several organic synthesis, it performs better role as a solvent (EU 2003). Iso-pentane is also used as a blowing agent, and neo-pentane is used in the manufacture of butyl rubber (Reference 1). Pentane can be released into the air from industrial production plants. It also finds use in the making of adhesives and glues.

Pentane, when released to the environment is expected to volatilize to the atmosphere, where it will undergo photochemical oxidation reactions with hydroxyl radicals (Reference 2).

There are certain characteristics of pentane which are highlighted below:

1. N-pentane has lower octane rating (RON of 61) rendering its utility as octane booster in gasoline/petrol. Higher branched alkanes are preferred as they gives RON of 91 and above. Therefore, it is nor added to petroleum products as an additive.
2. In the cold countries, the gasoline may contain in little over 1% v/v concentration so that its higher volatility aid in the ignition during sub-zero or cold conditions.

With respect to storage conditions for n-pentane, it is stored in cool, dry, ventilated areas to reduce fire and explosion risk (due to low boiling point, 36°C and high volatility, (Vap Pressure 514 mm Hg @ 25°C). The handling of pentane strictly requires the use of PPEs such as flame-resistant clothes, goggles and respiratory protection, besides, the use of non-sparking tools, removal of ignition sources and requiring isolated area. With respect to Fire safety, it comes under class A petroleum products. (Reference 3-4)

Inference: From the foregoing section, the referred literature and the standard practice of utilization of pentane, it can be inferred as follows,

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A) n-pentane has preferred usage as industrial chemical in variety of applications, such as blowing agent, aerosol propellant and industrial solvent in the manufacturing of specialty and variety of chemicals.
B) The use of n-pentane as an additive to the gasoline does not bring any added benefits as it has far lower octane number.
C) N-pentane comes under PESO guidelines necessitating to follow safe procedures for its storage and handling.

References:

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3. Carreón T. 2005. Aliphatic Hydrocarbons. In: Patty's Industrial Hygiene and Toxicology CDROM. Bingham E, B Cohrssen, CH Powell (ed.). John Wiley & Sons, Inc.
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